

An Efficient Cooperative Mission Planning Method of Heterogeneous Multi-USVs for Sea Area Coverage

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Abstract—Aiming at the problems of task time of each platform varies greatly, and the planned path takes too many turns for cooperative coverage mission of heterogeneous multi-USVs in irregular area, this paper presents an efficient cooperative mission planning method for heterogeneous multi-USVs in sea area coverage. Firstly, considering the difference of maneuver of each platform, the grid trading method is used to divide the task area reasonably, and then the backbone tree model is built in each area to generate the closed path, finally, the planning path is evolved to reduce the number of turns, and the path quality optimization of heterogeneous multi-USVs cooperative coverage in irregular areas are realized. To evaluate the degree of multi-USVs' collaboration for the area covering task, we design the indicator of Mission Time Volatility(MTV). The simulation results show that the method takes into account the speed difference and turn characteristics of each USV, and MTV of USVs' tasks is controlled within 4%. After the evolution, the number of turns is reduced by more than 25%, and the efficient cooperative coverage of heterogeneous multi-USVs is realized.

Keywords—area coverage, task planning, heterogeneous USVs, grid trading, backbone tree

I. INTRODUCTION

With the development of artificial intelligence (AI) technology, the application of AI technology in unmanned platform has been extended. The unmanned platform can carry a variety of loads with strong mobility and good autonomous mobility, has a distinct advantage in performing dangerous and tedious tasks. Among them, the USV(Unmanned Surface Vessel) is playing an increasingly important role in the sea mission, it is one of the important application directions to use multi-USVs for area cooperative patrol. At present, the path planning for the sea area coverage of multi-USVs is based on the basic geometry rules, although the planning speed is faster, the application scenario is limited to the regular mission area, and can not deal with the complex boundary task area and task scenario within the obstacles, and it can not allocate the sub-task area reasonably when the capability of heterogeneous USVs are different.

In recent years, the research on Multi-Agent Task Planning has made great progress. Area Coverage Path Planning (ACPP) is a sub-problem of Multi-Agent task planning. Compared with the Point-to-Point Path Planning (PPPP) problem, the main objectives of the area coverage path planning problem are full coverage, low repetition rate and fewer turns[1]. In some specific scenarios, ACPP is relatively sensitive to low repetition rate and low turn number, and the algorithm design has many constraints [2]. The algorithm of ACPP can be divided into cell

decomposition method, line scan segmentation method and grid method.

The cell decomposition method divides the task space according to the accessibility, simplifies the partition form as much as possible, and reduces the overlapping area as much as possible [3]. When the obstacles in the mission area are complex, the Boustrophedon cell decomposition method deals with convex polygons by choosing the right direction, and uses the round-trip walking mechanism to cover the region, it can reduce the number of subregions and make the final path shorter [4,5]. Relative to the cell decomposition method, Huang et al. proposed line scan segmentation method can effectively reduce the number of turns to achieve energy optimization [6]. The line scan segmentation method is improved by using ant colony optimization algorithm to find the area coverage planning which generates the minimum track length [7]. The optimization method proposed by Yu and Hung effectively reduces the number of turns in autonomous vehicle coverage path planning [8].

The common grid methods can be divided into two parts: the covering algorithm based on spanning tree and the covering algorithm based on biological heuristic neural network. The spanning tree coverage method uses the sensor of the agent to obtain the map data information, establish the grid map, and generate the final path [9]. On this basis, Feng and Katupitiya proposed a recursive algorithm based on minimum spanning tree to estimate the minimum number of agents required for area coverage with access frequency constraints [10]. Under the regional observation, an optimal path planning algorithm based on minimum spanning tree approximation algorithm and ant colony optimization was proposed by Akshya and Priyadarsini [11].

However, the above-mentioned studies' experiments are mostly based on the same-structure unmanned vehicle or unmanned aerial vehicle platforms, without considering the heterogeneous multi-USVs characteristics such as large turning radius, speed differences, etc. , so these can not be used in area covering task with heterogeneous USVs. In order to solve the above problems, this paper studies an efficient collaborative task planning method for heterogeneous multi-USVs to perform coverage missions in the sea area, divides the task area considering the speed difference of the platform. On this basis, the backbone tree method is used to carry out path planning, and through continuous iterative optimization, to improve the efficiency of multi-USVs coordination capability..

II. TASK DESCRIPTION

In this paper, the area coverage problem of heterogeneous multi-USVs in marine environment is summed up as the area sweeping coverage problem of multi-robot system with complex constraints. However, in a real experimental scenario, the following problems are encountered: firstly, the shape of the covered area is irregular; secondly, the initial position and speed of the USVs are different.

Considering the practical problems of ocean environment and covered area, and the different motion characteristics caused by platform heterogeneity, this paper proposes a new idea of region map division and path planning. Considering the different maneuverability of each USV, this paper uses the grid transaction method to divide the irregular task area, and allocates the sub-areas to the corresponding USV, in order to make the time required for each USV to cover their respective area approximately equal under ideal conditions. On this basis, the path planning of each sub-area is carried out by using the backbone tree model. At last, the backbone tree evolution method is used to plan a reasonable area coverage path for the corresponding USV in each sub-area, which reduces the turn times of the planning path and improves the availability of the initial scheme.

III. TASK AREA DIVIDE BASED ON GRID TRANSACTION METHOD

The division of the task area is realized by the task assignment-batch expansion-grid transaction, which makes the USV in the current sub-area as close to or within the operation area as possible, at the same time make the divided areas as uniform as possible. The regional division is divided into the following steps:

A. Map Pre-processing

The map pre-processing module is used to transform the parameterized map input into a set of grid maps for subsequent processing, the process is shown in Fig 1.

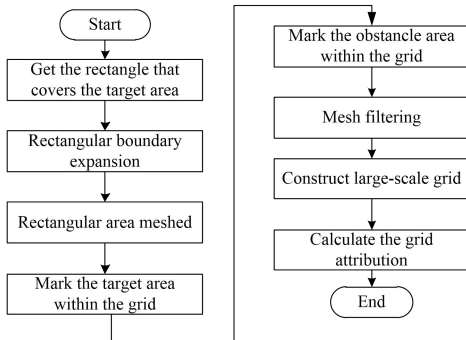


Fig. 1. Map pre-processing process

1) *Regional rasterization*: The rectangular area is divided into multiple grids.

2) *Construct large-scale grid*: Constructing large-scale grid by combining several grids to speed up area segmentation.

3) *Grid filtering*: Retaining the grids that constitute the passable area. Deleting the redundant grids and keeping the ones which are in the target area and not in the obstacle area. The retaining grids can constitute the area that the USV can pass through.

4) *Compute grid attribute*: Store the attributes of each grid and large-scale grid, including the effective neighbors of each grid, the boundary grids of each large-scale grid, and the adjacent large-scale grid.

B. Area Rough Segmentation

Area rough segmentation is used to divide the target area into several sub-areas and assign each sub-area to USV, which is located in or near the sub-area.

The implementation process includes the following two steps:

1) *The initial large-scale grids are specified for each USV*: The cost matrix is established by calculating the distance from the center point of each large scale grid to the current position of each USV. The problem can be modeled as a task assignment problem, that is, M large-scale grids are allocated to N USVs ($M > N$) with minimum total cost. The Hungarian algorithm is used to solve the optimal allocation scheme, and the initial large-scale grids are obtained for each of the N USVs to form N initial sub-areas.

2) *Through batch expansion*: each large-scale grid is allocated to an USV to complete the initial area division.

C. Area Fine Tuning

The result of area rough segmentation may have the problem that the time needed for each USV to cover the area may be quite different. Therefore, it is necessary to fine-tune the area of each sub-area, so that the time required for the completion of area coverage by each USV is as consistent as possible, thus speeding up the operation. In this paper, the regional division uniformity indicators is defined as:

$$m_i = (n_i s_g) / (2v_i r_i) \quad (1)$$

There, m_i is the indicators of the i th USV, n_i is the number of grids in the sub-area which covered by the i th USV, s_g is the area of single grid, v_i is the speed of the i th USV, r_i is the detection radius of the i th USV. Thus, the above indicators indicate the time needed for an USV to cover its area in an ideal situation.

Area fine tuning is achieved by following steps:

1) Fine-tuning strategy:

First, we calculate the number of grids that each sub-area would ideally contain, based on:

$$n_1 / (r_1 v_1) = n_2 / (r_2 v_2) \dots = n_k / (r_k v_k) = p \quad (2)$$

Hence:

$$p = (n_1 + n_2 + \dots + n_k) / (r_1 v_1 + r_2 v_2 + \dots + r_k v_k) \quad (3)$$

So for each sub-area, ideally the number of grids it contains is:

$$n_i = p r_i v_i \quad (4)$$

Defines the number of grids that the current subarea actually contains is n_i' ,

The grid migration scheme is formed as follows. for sub-area i, if $n_i^r > n_i$, traversing other sub-areas, if find sub-area j which $n_j^r < n_j$, we need to move the grids from sub-area i to sub-area j, the number of moved grids is :

$$n_{ij}^e = \begin{cases} n_i^r - n_i & n_i^r - n_i \leq n_j - n_j^r \\ n_j - n_j^r & n_i^r - n_i > n_j - n_j^r \end{cases} \quad (5)$$

Update n_i^r, n_j^r :

$$n_i^r = n_i^r - n_{ij}^e \quad (6)$$

$$n_j^r = n_j^r + n_{ij}^e \quad (7)$$

Through the above process, we can get the number of each sub-areas to transfer to other sub-areas, that is the fine-tuning scheme.

2) *Transfer the grid*: According to the fine-tuning scheme, each sub-area transfers a specific number of grids to the target sub-area. The process to complete the grid migration is shown in Fig 2.

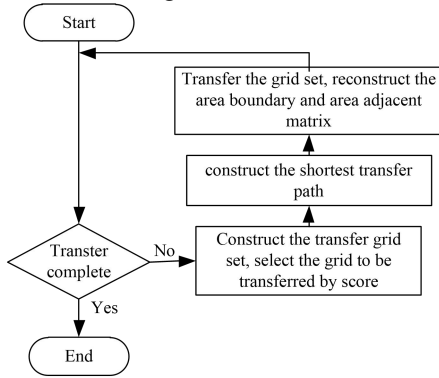


Fig. 2. Map pre-processing process

So far, the region segmentation processing is finished. In theory, we get a scheme which is suitable for the navigation capability of each USV and can cover all the regions with similar time-consuming.

IV. PATH PLANNING BASED ON BACKBONE TREE EVOLUTION

When an USV performs an area covering task, it hopes to generate a coverage path that can cover all parts of the target area without repeatedly visiting the same area, which means the ideal covering path should be a non-overlapping closed-loop path that can traverse the target area. In order to achieve this goal, this paper uses a backbone tree-based strategy to generate coverage paths.

The backbone tree-based path planning strategy is shown in Fig 3. In the target area, after the area is properly grid, the backbone tree model covering the target area can be built in this grid map. By tracing the backbone tree, a closed loop trajectory without traversing the target region and crossing is formed.

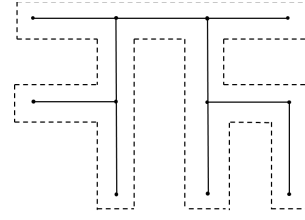


Fig. 3. Schematic diagram of path coverage strategy based on backbone tree

The steps for the backbone tree based policy to generate an overlay path are:

Firstly, grid the map, build a graph model connecting all the blocks of the current USV responsible area, and build any minimum spanning tree of this graph model, called its backbone tree.

By connecting the center points of each grid and forming the "outline" of the backbone tree around the backbone tree, a closed coverage path that can traverse all grids without repeated visits can be constructed.

The above is the region covering path generation method based on backbone tree, as shown in Fig 4. In the diagram, each color block represents an area for which the drone is responsible, where the solid line represents the backbone tree and the dashed line represents the coverage path planned around the backbone tree.

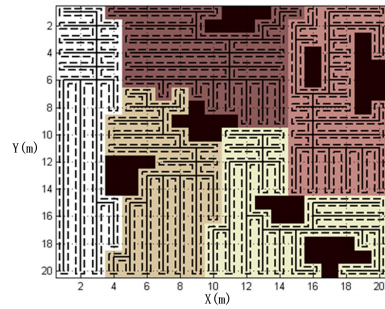


Fig. 4. Path planning method based on backbone tree

Since the USV does not have the ability to turn in place, and must maintain a certain speed during navigation to turn well, too many turns will seriously affect the quality of the path, so a good covering path should reduce the number of turns. When the backbone tree performs poor, the generated cover path may have a lot of turns, especially in the case of irregular boundaries and internal obstacles in the target area, the planning paths may be more chaotic. In this paper, the evolution strategy is used to optimize the backbone tree to improve the quality of the generated region covering path and reduce the turns of the path.

As mentioned earlier, one backbone tree is corresponded to a generated path, so the evolution of the path can be achieved through the evolution of the backbone tree. Firstly, a group of backbone trees is constructed by randomly selecting several roots as the initial population, and then iterative evolution is carried out. In each iteration, the backbone tree is optimized through introduction and reproduction mechanisms. The introduction mechanism randomly re-selects roots and builds backbone trees to avoid poor population quality caused by poor initial root selection, as shown in Fig 5.

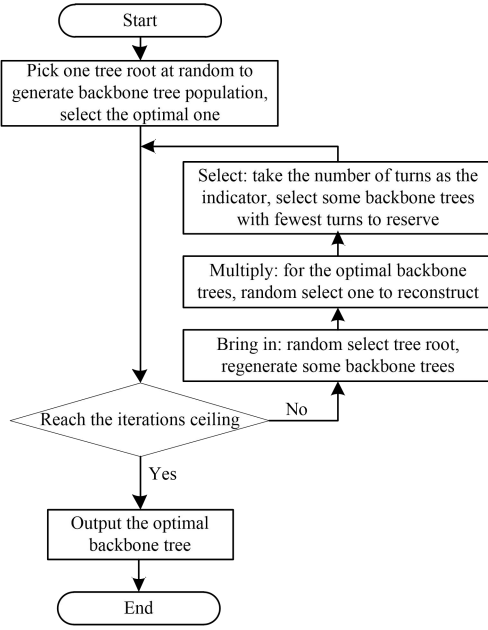


Fig. 5. Backbone tree evolution algorithm

The backbone tree is optimized by reconstructing a subtree through reproduction mechanism. The flow chart is shown in Fig 6.

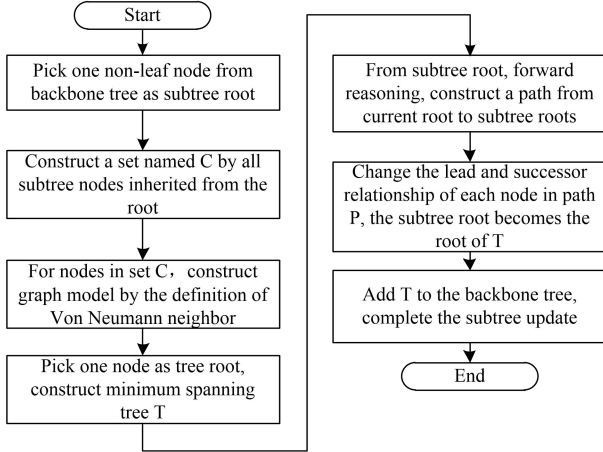


Fig. 6. Subtree optimization algorithm

Through the above mechanism, the backbone tree is iteratively optimized, and the optimal backbone tree is selected generation by generation, and the evolutionary optimization of patrol path is finally realized.

V. EXPERIMENTAL RESULTS AND ANALYSIS

In order to verify the task planning capability of the algorithm in different scenarios, scenarios such as obstacles in the task area, arbitrary shape task area, and variable number of agents were designed to verify the algorithm.

A. Obstacle Avoidance Ability Verification

In this experiments, the task scene was set as a hexagonal task area with obstacles, as shown in Fig 7. Four heterogeneous USVs with different speeds were dispatched to cover the area, and the path planning and evolution were carried out using our method, the comparison results of paths before and after evolution are shown in Fig 8. The left picture is the unoptimized path, and the right picture is the

planned path after evolution. Considering the actual application scenario, the path smoothing interpolation process has been carried out for the 90-degree turn. The vacant position in the figure is the obstacle area, the blue dots are the initial positions of the USVs, and the paths with great improvement are marked by different colors.

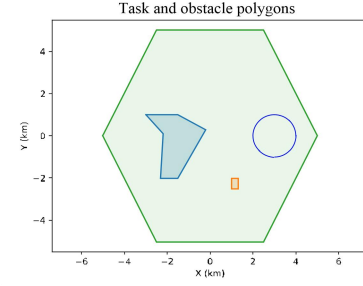


Fig. 7. Task and obstacle area

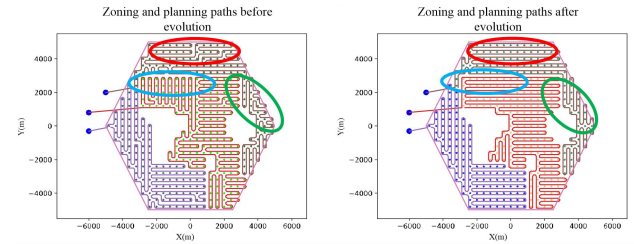


Fig. 8. Comparison of planned paths before and after evolution

It can be seen that the path planned by this method can achieve full path coverage in the region without path overlap while avoiding all obstacles, and the path quality is optimized after evolution.

B. Regional Adaptability Verification

In this section, three groups of task areas with different shapes are designed, all of which are irregular shapes, and the number of USVs is also set to 4. The method we proposed is adopted to conduct experiments on the three scenarios respectively, and the results are shown in Fig 9-11.

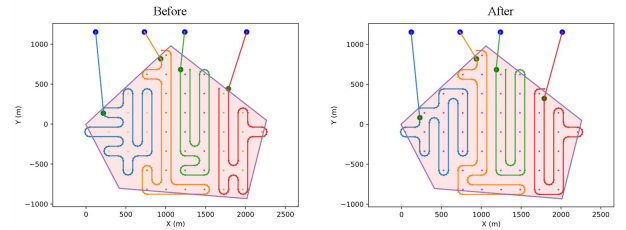


Fig. 9. Scenario 1

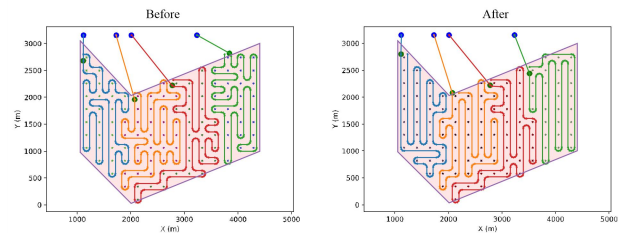


Fig. 10. Scenario 2

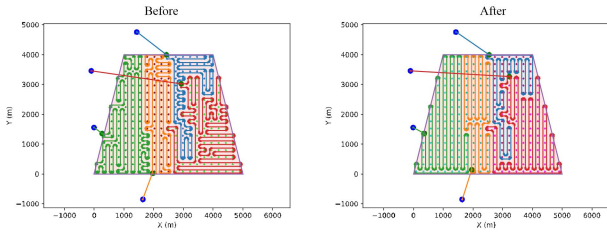


Fig. 11. Scenario 3

It can be seen from the figure that for any target region with irregular shape, the proposed method can plan a path covering the target region. To evaluate the uniformity of completion time for n USVs, we define the Mission Time Volatility(MTV) of USVs:

$$r = \frac{\sigma_t}{\mu_t} \times 100\% \quad (7)$$

Where, $\sigma_t = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$ is the standard deviation of

the task time of n USVs, and $\mu_t = \frac{1}{n}(x_1 + x_2 + \dots + x_n)$ is the average task time. The smaller the value of MTV, the better the division.

The path length and task consuming time of each USV planned in the above scenario are analyzed, and the results are shown in Table 1. It can be seen that in different mission area, although the path length of each USV is greatly different due to the difference of the velocity, the MTVs in different scenarios are all less than 4%.

TABLE I. COMPARISON OF PLANNING UNIFORMITY IN DIFFERENT SCENARIOS

Scenario ID	Area (km ²)	Ship ID	Speed (kn/h)	Path length(m)	Task time(s)	Average task time(s)	Length standard deviation(m)	Time standard deviation(s)	Mission time volatility(%)
1	2.88	1	3.1	6373	2055	1941	165	72	3.7
		2	3.3	6126	1856				
		3	3.1	5937	1915				
		4	3.3	6010	1938				
2	6.93	1	3.1	12748	4112	4314	593	128	3.0
		2	3.3	14370	4354				
		3	3.1	13413	4326				
		4	3.3	13841	4464				
3	16.00	1	3.3	22263	6746	6706	8348	174	2.6
		2	4.1	27279	6653				
		3	6.1	39472	6470				
		4	6.1	42433	6956				

The evolution effect of the three scenarios was analyzed, and the number of the turns of the planning paths before and after the evolution was compared. The number of turns was defined as one 90-degree turn before the path was smooth, and the 180-degree turn was considered to be composed of two 90-degree turns. The results are shown in Table 2. It can be seen that after the evolution of each scenario, the number of planned path turns is reduced by more than 25%. In the case of the same number of boats, the larger the task area, the more obvious the improvement effect.

TABLE II. COMPARISON OF PLANNING EFFECTS BEFORE AND AFTER EVOLUTION OF DIFFERENT SCENARIOS

Scenario ID	Number of initial scheme turns	Number of evolved scheme turns	Lifting ratio
1	70	52	25.7%
2	162	110	32.1%
3	248	158	36.3%

C. Platform Scale Adaptability Verification

In order to verify the supporting capability of the proposed method for different numbers of USVs, we carried out task planning in the task area of scenario 3 in the previous section under the condition that the number of boats was 3, 4 and 5, and the results were shown in Fig. 12, Fig. 11 and Fig. 13 respectively.

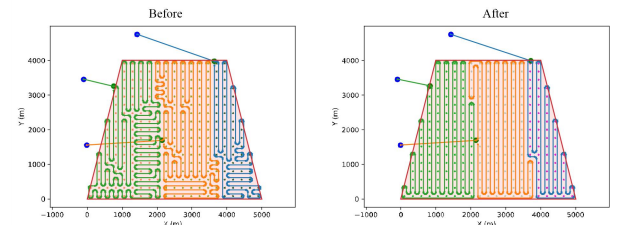


Fig. 12. Planning effect of 3 boats

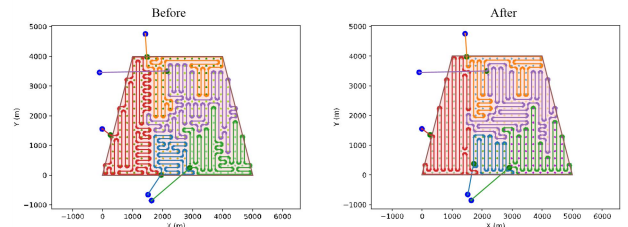


Fig. 13. Planning effect of 5 boats

As can be seen from the figure, for different numbers of USVs in the same target area, the proposed method can always plan a high quality path covering the target area. The path length and task time of each USV in the above scenarios are analyzed, and the results are shown in Table 3. The MTV in different scenarios is always less than 3%. This shows that the path planned by our method in different scenarios has

basically the same task completion time for each platform, and the efficient coordination of multi-USVs is realized.

TABLE III. COMPARISON OF PLANNING UNIFORMITY OF DIFFERENT NUMBER PLATFORMS

Scenario ID	Ship ID	Speed (kn/h)	Path length(m)	Task time(s)	Average task time(s)	Length standard deviation (m)	Time standard deviation(s)	Task time volatility(%)
1	1	3.3	29276	8871	8534	11034	239	2.6
	2	6.1	52129	8545				
	3	6.1	49935	8186				
2	1	3.3	22263	6746	6706	8348	174	2.6
	2	4.1	27279	6653				
	3	6.1	39472	6470				
	4	6.1	42433	6956				
3	1	2.1	13283	6325	6134	9153	151	2.5
	2	3.1	20102	6091				
	3	4.1	25705	6269				
	4	6.1	35980	5898				
	5	6.1	37120	6085				

The evolution effects of all the three scenarios were analyzed, and the turning times of the planning scheme before and after the evolution were compared. The results are shown in Table 4. It can be seen that the turning times of the planned paths after the evolution of each scenario are all reduced by more than 30%. In the case of the same task area, the fewer the number of boats, the better the path optimization.

TABLE IV. COMPARISON OF PLANNING EFFECTS BEFORE AND AFTER EVOLUTION OF DIFFERENT NUMBER OF PLATFORMS

Scenario ID	Ship number	Number of initial scheme turns	Number of evolved scheme turns	Lifting ratio
1	3	238	94	60.5%
2	4	248	158	36.3%
3	5	316	220	30.4%

To sum up, aiming at the multi-USVs area covering task we part the solution into two stages. In first stage, we proposed a kind of region division method, which can be used in maps of irregular shapes containing multiple obstacles. With the method, task areas can be assigned to each USV reasonably in according of platform number and each platform's velocity. This ensures that the completion time of tasks for each platform is essentially consistent. In the second stage, we also design the backbone tree path-planning method to generate obstacle-free and conflict-free paths for each platform. The backbone tree path-planning method can also evolve the results by adjusting the paths to reduce the number of turns. By combining the region segmentation method and the backbone tree path-planning method, we have achieved efficient path planning for heterogeneous multi-USVs.

VI. CONCLUSION

In order to solve the problem of collaborative area coverage of heterogeneous multi-USVs, this paper proposes a method of area division based on grid transactions and path optimization based on backbone tree evolution. We also conducted simulation experiments under different conditions, including the presence of obstacles, various irregular shapes of map areas, and different numbers of USVs. The results showed that, our method can effectively solve the problem. To evaluate the degree of multi-USVs' collaboration for the

area covering task, we design the indicator of Mission Time Volatility. Using our method, the MTV of USVs' tasks is less than 4% and the number of turning times of the planned path has been reduced by more than 25% after evolving the planned path, which ensured the efficient collaboration among multiple USVs in the area covering task.

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